WiFi-based Global Localization in Large-Scale Environments Leveraging Structural Priors from osmAG

Xu Ma[1], Jiajie Zhang[1], Fujing Xie[1] and Sören Schwertfeger[1]

***Abstract*—Global localization enables a robot to determine its absolute position in a known map, forming the basis for long-term autonomy and addressing the kidnapped robot problem in GPS-denied environments such as indoor and underground facilities. We propose a novel WiFi-based localization framework that leverages ubiquitous wireless infrastructure and the OpenStreetMap Area Graph (osmAG) for large-scale indoor environments. Our approach integrates structure-aware signal propagation modeling with osmAG's geometric and topological priors. In the offline phase, an iterative optimization algorithm localizes unknown WiFi Access Points (APs) by explicitly modeling structural elements (e.g., walls, doors) attenuation, achieving a mean localization error of 3.79 m (35.3% improvement over geometry-unaware trilateration) and embeds these optimized APs into the osmAG map as semantic nodes. In the online phase, real-time robot localization utilizes this augmented osmAG map, yielding a mean error of 3.12 m in fingerprinted areas and 3.83 m in unfingerprinted areas. Extensive benchmarking against four baselines—KNN fingerprinting, Gaussian Process Path Loss (GP-PL), probabilistic fusion (GPoE), and enhanced fingerprinting with metaheuristic optimization (IRPLS-ABCL)—demonstrates that our approach achieves superior generalization. While all baselines degrade to 11–20 m error in unfingerprinted areas, our framework maintains high accuracy. Furthermore, our map representation is orders of magnitude more space-efficient than dense fingerprint databases (ours: 44 KB vs fingerprinting: 11068 KB). Validated across a complex 11,025 m^2 multi-floor environment, this framework presents a scalable, robust solution for the kidnapped robot problem. The code and dataset are available at https://anonymous.4open.science/r/osmag-wifi-localization**

I. INTRODUCTION

Global localization is the cornerstone of long-term robotic autonomy, particularly for addressing the *kidnapped robot problem* in GPS-denied environments. While LiDAR and visual approaches set the benchmark for accuracy, they face significant barriers regarding hardware costs, computational load, and vulnerability to perceptual aliasing in texture-less or dynamic scenarios. These limitations necessitate cost-effective, lightweight alternatives capable of providing robust global priors.

WiFi-based localization [1] emerges as a compelling solution by capitalizing on ubiquitous wireless infrastructure. However, existing methodologies face critical limitations. *Fingerprinting approaches* [2], including enhanced variants with interval radio-map analysis [3], depend on exhaustive manual surveys and suffer from the *extrapolation problem*: failing to generalize to unvisited areas outside their training distribution, as illustrated in Fig. 1 (top). *Probabilistic methods* based on Gaussian Processes [4], [5] offer spatial interpolation, yet still assume smooth signal fields and cannot explicitly model wall-induced discontinuities. Traditional *model-based methods* [6] further oversimplify propagation by ignoring complex environmental geometries.

[1]The authors are with the Key Laboratory of Intelligent Perception and Human-Machine Collaboration - ShanghaiTech University, Ministry of Education, China. {maxu2023, zhangjj2023, xiefj, soerensch}@shanghaitech.edu.cn

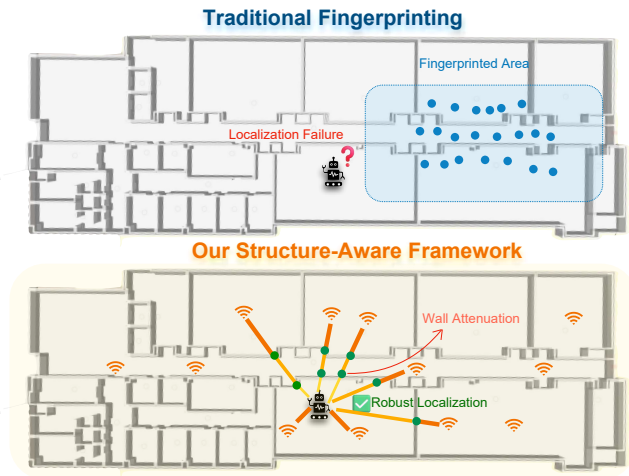


Fig. 1: **Overcoming the extrapolation barrier in WiFi localization. Top:** Traditional fingerprinting methods rely on dense historical data (blue dots), leading to localization failure when the robot enters unfingerprinted areas (red question mark). **Bottom:** Our proposed framework leverages *osmAG* [7] structural priors to explicitly model structure-aware signal propagation. By computing wall attenuation at intersection points (green nodes), the model dynamically adjusts signal strength estimates (visualized by the thinning rays), enabling robust global localization even in areas devoid of fingerprint data.

Our work bridges this gap by proposing a structure-aware framework that explicitly models Radio Frequency (RF) attenuation caused by physical structures (see Fig. 1, bottom). The success of such a model hinges on a map representation that rigorously defines the environment's geometric structure. To this end, we leverage the *OpenStreetMap Area Graph (osmAG)* [7], detailed in subsection III-A.

By fusing osmAG's structural priors with a deterministic signal model, we introduce a localization system that localizes unknown WiFi Access Points (APs) in an offline phase and enables real-time robot localization in an online phase using the augmented osmAG map. The primary contributions are:

- **Structure-Aware AP Mapping:** A novel calibration framework that fuses signal propagation models with osmAG structural priors, significantly reducing the need

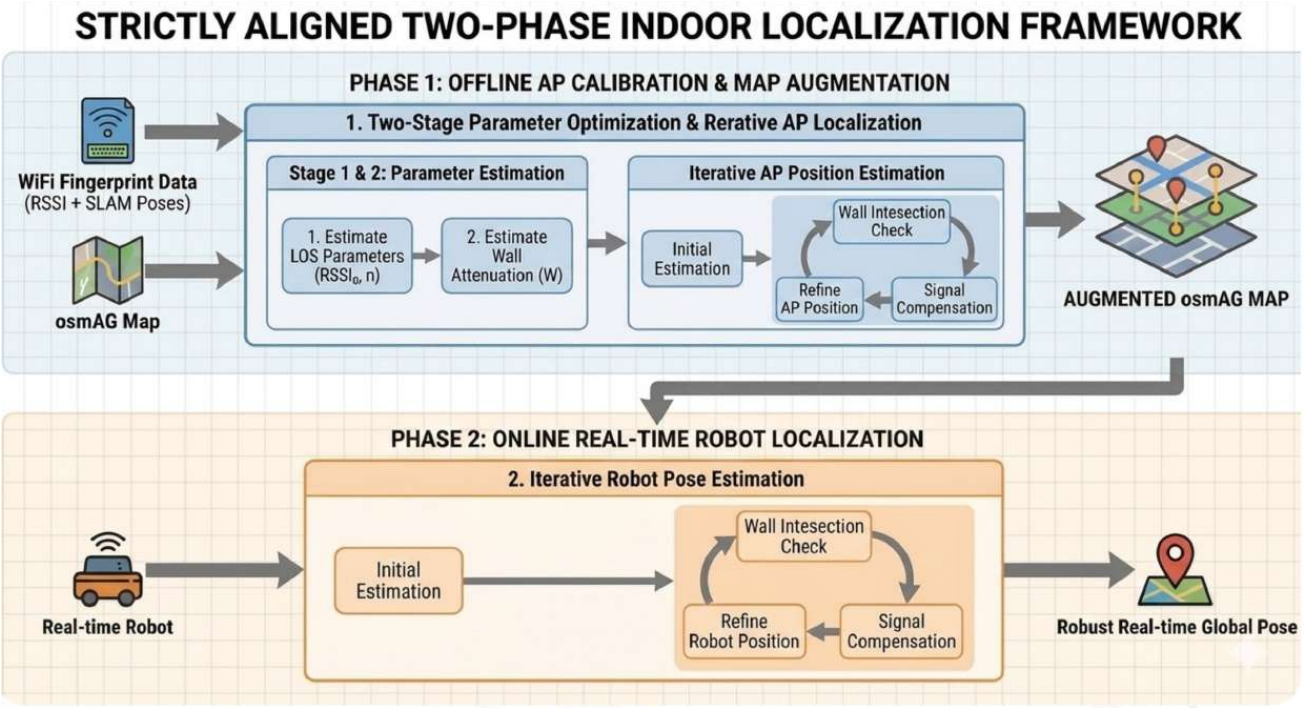


Fig. 2: The proposed structure-aware localization pipeline. Offline Calibration: The system ingests raw WiFi scans and SLAM poses to inversely solve for latent environmental parameters (AP locations $\mathbf{p}_{AP}$, path loss exponent n^*, and average wall attenuation $\overline{W}$) via an iterative optimization loop. Augmented osmAG map: The resolved physical parameters and AP locations are permanently injected into the map as semantic tags. Online Localization: The robot retrieves these structural priors to perform real-time ray casting and signal prediction, estimating its global pose.

for exhaustive manual AP surveying.

- **Robust Generalization & Efficiency:** A localization system that significantly outperforms KNN fingerprinting [2], Gaussian Process (GP-PL) [4], probabilistic fusion (GPoE) [5], and enhanced fingerprinting with meta-heuristic optimization (IRPLS-ABCL) [3]. In unfingerprinted areas (out-of-distribution), our method achieves a mean error of 3.83 m versus 11.00–20.21 m for all baselines, while reducing WiFi-related map storage by orders of magnitude compared to dense fingerprint databases.
- **Scale Validation & Open Resources:** Validated across a complex 11,025 m^2 multi-floor environment, our method demonstrates consistent accuracy across diverse topologies, from narrow corridors to open atriums. The code and dataset are available at https://anonymous.4open.science/r/osmag-wifi-localization.

II. RELATED WORK

This section reviews global localization methodologies, transitioning from sensor-heavy approaches to RF-based solutions, and positions our structure-aware framework within this landscape.

A. Sensor-heavy Global Localization

Visual and LiDAR approaches represent the current benchmark for accuracy but face deployment challenges. Visual localization, ranging from Bag-of-Words (BoW) [8] to deep global descriptors like NetVLAD [9], degrades significantly in texture-less indoor scenarios and suffers from perceptual aliasing under lighting variations [10]. Similarly, LiDAR-based methods [11], [12] leverage precise 3D geometry but encounter *geometric ambiguity* in symmetrical environments (e.g., long corridors) and impose high computational costs for global registration. These limitations necessitate cost-effective, lightweight alternatives like WiFi for robust redundancy.

B. WiFi-based Localization

1) Fingerprinting & Enhanced Fingerprinting: Classic fingerprinting methods [2], [13] construct dense RSSI databases for k-nearest-neighbor matching. While effective in surveyed areas, they are labor-intensive and degrade rapidly in unseen regions. To improve radio-map quality, Zhang et al. [3] proposed IRPLS-ABCL, which enhances the offline fingerprint database via an Interval Random Parameter Lognormal Shadowing model and refines positions online with Adaptive Bayesian Comprehensive Learning PSO (ABCL-PSO). This pipeline improves fingerprint fidelity through interval analysis of propagation parameters but still relies on globally fitted models without explicit structural awareness.

2) Probabilistic & Model-based Approaches: Gaussian Process (GP) methods [14] model signal strength as continuous spatial functions to reduce survey costs. Miyagusuku et

al. [4] incorporate path-loss propagation models as informative GP priors (GP-PL), fitting a per-AP log-distance model and training a GP on the residuals with variance bounding, achieving improved interpolation in sparse regimes. The generalized Product of Experts (GPoE) [5] further fuses per-AP posteriors via information-gain/entropy-based weights. However, these probabilistic methods assume signal smoothness and fail to model discrete discontinuities caused by physical obstacles such as walls.

In contrast, our framework explicitly integrates *osmAG* structural priors into a deterministic signal model. Unlike fingerprinting methods that cannot extrapolate, or GP-based models that smooth over obstacles, we leverage precise wall geometry for dynamic ray-casting, shifting the paradigm from "pattern matching" to "physical modeling."

III. METHOD

Our proposed large-scale indoor localization framework is architecturally divided into two primary phases: an *offline AP calibration phase* and an *online robot localization phase*. As shown in Fig. 2, in the offline phase, we utilize a calibration dataset of time-synchronized WiFi RSSI measurements and robot poses (obtained via LIO-SAM [15]). An optimization algorithm ingests this dataset alongside *osmAG* wall geometry to jointly recover AP locations and attenuation parameters, which are then embedded into the map.

In the online phase, the robot queries this augmented map and fuses the retrieved priors with live RSSI scans for real-time global localization.

A. osmAG Map Representation

Our framework leverages the OpenStreetMap Area Graph (osmAG) [7] to establish a unified representation of both environmental geometry and wireless signal data. osmAG stores map data in XML format, accepting input from CAD drawings, 2D grid maps, or 3D point cloud maps [7]. Unlike conventional occupancy grids, osmAG utilizes a semantic graph structure built from two primitives: *nodes*—discrete coordinate points representing key locations (e.g., room corners, doorways)—and *ways*—ordered sequences of nodes that form polygons delineating structural elements such as walls, corridors, and rooms. Together, these primitives encode both the *geometry* (shape and size of spaces) and the *topology* (connectivity between spaces, e.g., how corridors link rooms), which is essential for the ray-wall intersection checks in our structure-aware model.

We further extend the osmAG schema using its flexible *tagging* system to embed WiFi data directly into the map structure. This capability serves two distinct phases of our pipeline:

1) **Fingerprint Data (Offline Input):** For the calibration phase, raw RSSI measurements and synchronized robot poses are stored as semantic nodes. This allows the osmAG file itself to serve as a self-contained training dataset.

Fig. 3: Visualization of the augmented *osmAG* map representation integrating physical geometry with WiFi signal parameters. The grey structures depict the 3D layout of walls and rooms, while orange nodes represent the optimized locations of WiFi Access Points (APs). As detailed in the callout box, each AP is stored as a semantic node containing key-value pairs (tags) for essential properties such as BSSID, floor level, and physical coordinates. This unified representation allows the robot to query both structural constraints and signal priors directly from a single map file.

2) **AP Parameters (Online Prior):** For the online phase, the optimized 3D locations and attributes of APs are permanently injected into the map. Fig. 3 visualizes this augmented representation, where APs are stored as semantic nodes containing key-value pairs (e.g., BSSID, floor level) alongside the physical geometry.

By consolidating structural constraints and signal priors into a single XML file, the robot can query both wall attenuation factors and AP locations from a unified source.

B. Offline Phase: AP Localization

The primary objective of the offline phase is to localize all Access Points (APs) from previously collected fingerprint data and embed their positions into the osmAG map, thereby establishing the foundational data layer required for the subsequent online robot localization.

1) Structure-Aware Signal Propagation Model: In this stage, a small subset of APs with known ground-truth positions are used as anchor points to learn the global model parameters (see Section IV-B for specific experimental settings).

We adopt the log-distance path loss model, a standard for describing indoor radio wave propagation. The model relates the received signal strength to the distance between a transmitter and a receiver:

$$\text{RSSI} = \text{RSSI}_0 - 10n\log_{10}(d) - \sum_k W_k + X_\sigma \qquad (1)$$

where RSSI_0 is the reference signal strength at 1 m, n is the path loss exponent, d is the 3D Euclidean distance between the AP and the robot, W_k is the attenuation factor for the k-th structure elements, k is the number of structure elements, and X_σ is a zero-mean Gaussian random variable

representing shadowing effects. The unknown parameters to be estimated are the reference signal strength RSSI_0, the path loss exponent n, and the wall attenuation factors W_k. To ensure computational tractability, we simplify the wall model to estimate a single average attenuation factor $\overline{W}$ for all structural elements. The following two-stage optimization procedure is designed to sequentially recover these parameters.

2) Two-Stage Parameter Optimization: To accurately estimate the model parameters, we employ a two-stage optimization process that leverages the geometric information from the osmAG map.

1) **Stage 1: LOS Parameter Estimation.** We first isolate a set of Line-of-Sight (LOS) measurements. For each anchor AP-Robot pair, we perform a geometric intersection test between the direct signal path and the polygons stored in the osmAG map. If no intersection occurs, the measurement is classified as LOS. Using this LOS dataset ($\mathcal{S}_{\text{LOS}}$), we solve for the baseline parameters (RSSI_0^*, n^*) via least-squares regression:

$$(\text{RSSI}_0^*, n^*) = \arg\min_{\text{RSSI}_0, n} \sum_{(i,j)\in\mathcal{S}_{\text{LOS}}} e_{ij}^2 \tag{2}$$

$$e_{ij} = \text{RSSI}_{ij} - \text{RSSI}_0 + 10n\log_{10}(d_{ij}) \tag{3}$$

where (i, j) represents the measurement pair between the i-th anchor AP and the j-th robot position, RSSI_{ij} is the measured signal strength, and d_{ij} is the 3D Euclidean distance.

2) **Stage 2: NLOS Wall Attenuation Estimation.** Signals whose direct path is obstructed by one or more walls are classified as Non-Line-of-Sight (NLOS). With the baseline parameters fixed, we estimate the effective wall attenuation factor $\overline{W}$ by minimizing the residual error for all NLOS measurements:

$$\overline{W} = \arg\min_{W} \sum_{(i,j)\in\mathcal{S}_{\text{NLOS}}} e_{ij}^2 \tag{4}$$

$$e_{ij} = \text{RSSI}_{ij} - \text{RSSI}_0^* + 10n^*\log_{10}(d_{ij}) + N_{ij}W \tag{5}$$

where N_{ij} is the number of walls intersecting the path between measurement i and j.

3) Iterative AP Position Estimation: With the environmental parameters $(\text{RSSI}_0^*, n^*, \overline{W})$ established, we proceed to estimate the unknown position of APs using an iterative optimization approach, summarized in Algorithm 1.

The core idea is a *Virtual LOS* compensation strategy. The initialization (Stage 1 in Algorithm 1) assumes all measurements are LOS, converting RSSI directly to distance via the learned path loss model. Because wall attenuation is unmodeled at this stage, NLOS signals appear weaker than expected, causing the solver to overestimate distances and pull the initial AP estimate away from its true location (Fig. 4a).

The iterative refinement (Stage 2) corrects this bias by leveraging osmAG geometry. At each iteration, the algorithm performs ray-wall intersection tests from the current AP estimate to all robot positions and compensates each NLOS measurement by adding $N_i \cdot \overline{W}$ back to the observed RSSI—effectively projecting it onto a virtual LOS condition. The corrected distances yield a tighter trilateration, which in turn updates the AP estimate and changes the wall intersection counts for the next iteration. This alternating process converges because each wall compensation step reduces the systematic distance overestimation, monotonically shrinking the residual until the position update falls below a threshold ϵ. Fig. 4 illustrates this process, showing how the algorithm converges to a solution consistent with both signal measurements and physical geometry.

Algorithm 1 Structure-Aware Iterative AP Localization

Require: WiFi Calibration Dataset $\mathcal{D} = \{(\mathbf{p}_i, \text{RSSI}_i)\}_{i=1}^M$, where $\mathbf{p}_i$ is the robot pose; *osmAG* map structure $\mathcal{M}$ (wall polygons); Learned Global Parameters $\Theta = (\text{RSSI}_0^*, n^*, \overline{W})$.

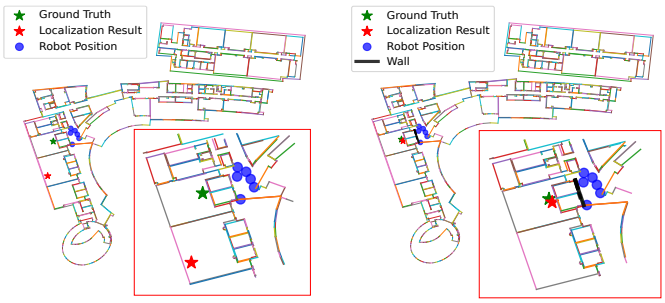
Ensure: Estimated 3D position of the Access Point $\mathbf{p}_{AP}$.

Stage 1: Initialization (Geometry-Unaware)
for each measurement $i \in \{1, \ldots, M\}$ **do**
  Estimate distance assuming LOS:
  $\hat{d}_i^{(0)} \leftarrow 10^{(\text{RSSI}_0^* - \text{RSSI}_i)/(10n^*)}$
end for
Solve initial position via least-squares trilateration:
$\mathbf{p}_{AP}^{(0)} \leftarrow \arg\min_{\mathbf{p}} \sum_{i=1}^M \left(\|\mathbf{p} - \mathbf{p}_i\| - \hat{d}_i^{(0)}\right)^2$
Stage 2: Iterative Refinement (Structure-Aware)
for $t \leftarrow 1$ to T_{max} **do**
  for each measurement $i \in \{1, \ldots, M\}$ **do**
    Construct signal ray $\mathcal{L}_i \leftarrow \text{LineString}(\mathbf{p}_{AP}^{(t-1)}, \mathbf{p}_i)$
    Detect wall intersections: $N_i \leftarrow \text{Intersects}(\mathcal{L}_i, \mathcal{M})$
    if $N_i > 0$ **then** ▷ NLOS Detected
      Compensate signal attenuation (Virtual LOS):
      $\text{RSSI}_i' \leftarrow \text{RSSI}_i + N_i \cdot \overline{W}$
    else ▷ LOS Preserved
      $\text{RSSI}_i' \leftarrow \text{RSSI}_i$
    end if
    Update distance estimate using compensated RSSI:
    $\hat{d}_i^{(t)} \leftarrow 10^{(\text{RSSI}_0^* - \text{RSSI}_i')/(10n^*)}$
  end for
  Solve refined position with updated distances:
  $\mathbf{p}_{AP}^{(t)} \leftarrow \arg\min_{\mathbf{p}} \sum_{i=1}^M \left(\|\mathbf{p} - \mathbf{p}_i\| - \hat{d}_i^{(t)}\right)^2$
  Check Convergence:
  if $\|\mathbf{p}_{AP}^{(t)} - \mathbf{p}_{AP}^{(t-1)}\| < \epsilon$ **then**
    break
  end if
end for
return Refined AP Position $\mathbf{p}_{AP}^{(t)}$

C. Online Phase: Real-Time Robot Localization

In the online phase, the robot uses the augmented osmAG map to determine its own position.

1) Real-Time Data Acquisition: The robot continuously scans for WiFi signals, collecting RSSI measurements over a



(a) Initial position estimate (b) Final position estimate

Fig. 4: Iterative AP localization for a single Access Point. (a) Initial estimate using a log-distance model, showing significant deviation from the ground truth (green star). (b) Final estimate after iterative optimization with wall attenuation, converging closer to the ground truth after 10 iterations. Blue points represent robot positions, and the red box highlights a zoomed-in view of the final estimate.

brief temporal window (e.g., 3-5 seconds) and averaging them to ensure signal stability.

2) Iterative Position Estimation: The robot's position is estimated using an analogous iterative framework.

1) **Initialization:** An initial robot position estimate $\mathbf{p}_\mathrm{r}^{(0)}$ is computed via trilateration using detected APs and their map locations.
2) **Iterative Refinement:** At each iteration t, the system performs wall intersection checks between the current robot estimate $\mathbf{p}_\mathrm{r}^{(t-1)}$ and all visible APs. Compensated distances $\hat{d}_k^{(t)}$ are computed analogously to Algorithm 1 (using ray-wall intersections between the current robot estimate and each visible AP). It then minimizes the residual:

$$\mathbf{p}_\mathrm{r}^{(t)} = \arg\min_{\mathbf{p}} \sum_{k=1}^{K} \left(\|\mathbf{p} - \mathbf{p}_k^{\mathrm{AP}}\| - \hat{d}_k^{(t)} \right)^2 \tag{6}$$

where $\mathbf{p}_k^{\mathrm{AP}}$ is the known position of the k-th visible AP. The process terminates when the position update falls below a threshold (ϵ) or T_{max} is reached.

IV. EXPERIMENTS

This section evaluates both the AP localization accuracy and the online robot localization performance of our framework.

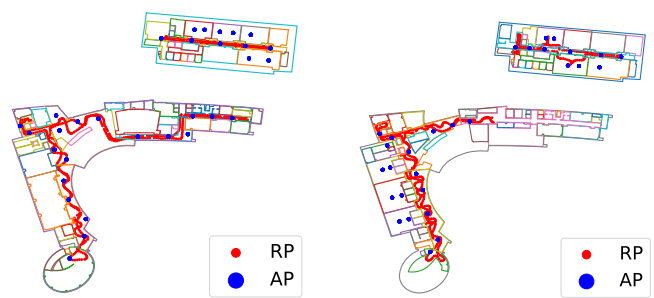
A. Experimental Setup

Our experiments were conducted in a large-scale multi-floor indoor environment at a university campus, covering 11,025 m^2 across two distinct buildings, as illustrated in Fig. 5. To comprehensively evaluate the robustness of our approach under varying spatial constraints, we selected two contrasting environments:

1) **Building A**, characterized by *narrow corridors* and dense partition walls, representing complex, high-attenuation environments.
2) **Building B**, featuring *wide open spaces* and expansive atriums, representing scenarios with sparse structural constraints.

This topological diversity ensures that our structure-aware model is rigorously validated against distinct signal propagation conditions, ranging from heavy obstruction to line-of-sight dominance. The environment comprises classrooms, labs, and offices with various wall types (concrete, drywall, glass). We used an Agilex Hunter SE mobile platform equipped with an EDUP EP-AC1681 WiFi receiver and a Hesai 64-channel LiDAR for data acquisition. LIO-SAM [15] was used to obtain ground truth robot positions, while 48 AP ground truths were manually surveyed.

The dataset is partitioned as follows: $K = 8$ APs with known positions serve as anchors for parameter learning (Section IV-B); the remaining 40 APs are used to evaluate the offline AP localization (Section IV-C). For online robot localization, 66 positions within fingerprinted areas form the in-distribution test set, while 7 positions in unfingerprinted areas constitute the out-of-distribution (OOD) test set (see Fig. 7).



(a) Floor 1 (b) Floor 2

Fig. 5: osmAG layout of the experimental environment across Building A and Building B. (a) Floor 1 and (b) Floor 2 show Robot Positions (RP) and ground truth Access Points (APs).

B. Parameter Learning

To instantiate our structure-aware signal propagation model, we executed the **Two-Stage Parameter Optimization** pipeline described in Section III-B2. We manually surveyed $K = 8$ APs as anchor points to learn the global model parameters. These 8 anchors are *excluded* from the subsequent AP localization evaluation (reported only over the remaining 40 APs in Section IV-C below). We selected the first floor of Building A as our training area, which is representative of the complex environment where the framework is to be deployed. By minimizing the objective functions defined in Eq. 2 and Eq. 4, the optimization yielded the following global physical parameters:

- Reference Signal Strength (RSSI_0^*): -28.79 dBm
- Path Loss Exponent (n^*): 2.5
- Wall Attenuation Factor ($\overline{W}$): 10.77 dB

TABLE I: Quantitative Comparison of Robot Localization Performance Across Different Scenarios.

Method	**Fingerprinted Areas** (In-Distribution)				**Unfingerprinted Areas** (Out-of-Distribution, N=7)			
	Mean (m)	**Std.** (m)	**RMSE** (m)	**95%** (m)	**Mean** (m)	**Std.** (m)	**RMSE** (m)	**95%** (m)
KNN Fingerprinting	3.42	2.50	4.24	7.85	20.21	9.48	22.32	36.18
GP-PL [4]	3.18	2.81	3.80	6.50	13.70	11.50	14.80	22.90
GPoE [5]	3.12	1.77	3.59	6.63	14.71	2.67	14.95	18.72
IRPLS-ABCL [3]	**3.05**	1.88	3.59	6.63	11.00	2.41	11.26	13.65
Ours (osmAG-Structure)	3.12	**1.35**	**3.34**	**4.65**	**3.83**	**0.78**	**3.91**	**4.72**

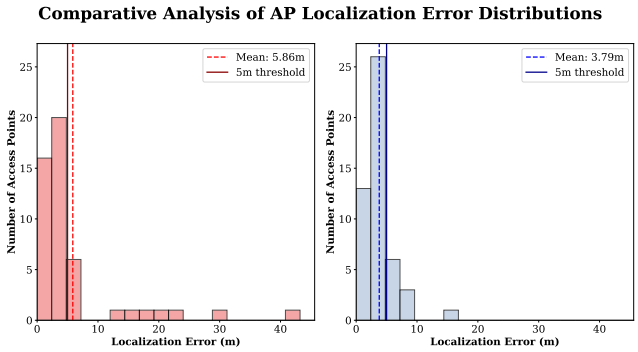


Fig. 6: Per-AP localization error comparison between the geometry-unaware trilateration baseline and our structure-aware iterative optimization over 40 unknown APs. Our method consistently reduces error and exhibits a tighter distribution.

These learned parameters were then fixed and applied consistently throughout all subsequent experiments for both the offline AP localization and online robot localization phases.

C. Access Point Localization

To validate the offline calibration phase, we benchmarked our structure-aware AP estimation against a geometry-unaware trilateration baseline. The baseline, which ignores environmental constraints, yielded a mean localization error of 5.86 m with a high standard deviation (7.82 m), reflecting its susceptibility to NLOS effects. In contrast, our iterative optimization leveraging osmAG structural priors significantly refined these estimates, reducing the mean error to **3.79 m** (std. 2.52 m) over the **remaining 40 unknown APs**. This **35.3% reduction in error**, combined with the visibly tighter error distribution shown in Fig. 6, demonstrates that explicitly modeling wall attenuation is crucial for recovering accurate AP positions from unlabeled data.

D. Robot Localization

To evaluate both the localization accuracy and the *generalization capability* of our system, we benchmarked against four state-of-the-art baselines:

1) **KNN Fingerprinting** [2]: Dense database matching without any physical model.
2) **GP-PL** [4]: GP with path-loss priors; assumes spatially smooth residual fields.
3) **GPoE** [5]: Multi-AP GP fusion; captures spatial continuity but ignores wall-induced discontinuities.
4) **IRPLS-ABCL** [3]: Enhanced radio map with interval analysis and PSO refinement; lacks explicit structural awareness.

These baselines span dense matching (KNN), spatial smoothing (GP-PL, GPoE), and radio-map enhancement (IRPLS-ABCL); see Section II-B for detailed descriptions. The quantitative results are summarized in Table I.

1) Localization within Fingerprinted Areas: In the fingerprinted areas (In-Distribution), all methods except KNN achieve comparable mean errors around 3 m, confirming that data-rich regions are well-served by existing approaches. Our method achieves a mean error of 3.12 m with the tightest spread (std. 1.35 m, 95% = 4.65 m). IRPLS-ABCL (3.05 m) and GPoE (3.12 m) attain similar means, while GP-PL (3.18 m) and KNN (3.42 m) remain competitive. The key differentiator is variance: our method yields the lowest standard deviation (1.35 m) and the tightest 95th-percentile bound (4.65 m), confirming that osmAG structural priors provide effective constraints even where data-driven approaches already perform well.

2) Generalization to Unfingerprinted Areas: The critical advantage emerges in unfingerprinted areas. **KNN** failed most severely (20.21 m), having no mechanism to extrapolate beyond observed reference points. **GP-PL** (13.70 m, std. 11.50 m) and **GPoE** (14.71 m) both suffered from wall-induced signal discontinuities that violate their smoothness assumptions. **IRPLS-ABCL** achieved a lower mean (11.00 m) with moderate variance (std. 2.41 m), yet still lacks explicit structural reasoning. In contrast, our framework maintained **3.83 m** mean error (std. 0.78 m, 95% = 4.72 m) by leveraging osmAG ray-casting to predict wall-induced attenuation in unvisited areas, demonstrating the robustness required for large-scale deployment without exhaustive fingerprinting. We note that the OOD test set comprises 7 points deliberately selected to span the most topologically diverse and challenging unseen conditions in our environment. Despite the limited sample size, the observed performance gap is substantial (3.83 m vs. 11.00–20.21 m for all baselines), representing a margin that far exceeds the within-method variance, lending confidence to the conclusion even under small-N conditions.

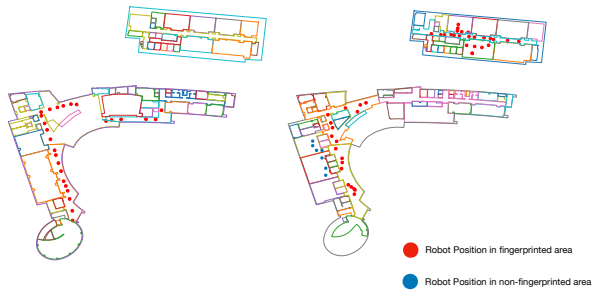


Fig. 7: Spatial distribution of ground truth robot positions used for localization evaluation across two floors. The dataset is split into two testing scenarios: **Red points** ($N = 66$) represent test samples within fingerprinted areas (In-Distribution), used to benchmark standard accuracy. **Blue points** ($N = 7$, specifically selected from the most challenging unseen topologies including corridors crossing buildings and open atrium without prior scans) highlight unfingerprinted areas (Out-of-Distribution, OOD) to validate the system's extrapolation capability using osmAG structural priors.

E. Storage Efficiency

Beyond localization accuracy, our map representation offers orders-of-magnitude improvements in storage efficiency compared to traditional fingerprinting databases. Table II summarizes the comparison.

TABLE II: Storage Efficiency Comparison of WiFi Map Representations.

Representation	Points	Size (KB)	Per-Unit
Fingerprint DB	1,491 scans	11,068	7.42 KB/scan
osmAG + AP (Ours)	48 APs	44	<1 KB/AP

Our augmented osmAG representation achieves a **compression ratio of 251:1** for WiFi-related data compared to the raw fingerprint database, while the base osmAG map size remains unchanged. This enables the deployment of large-scale semantic maps on resource-constrained mobile robots without the overhead of dense database maintenance.

V. CONCLUSIONS

We have presented a structure-aware global localization framework that replaces dense data requirements with explicit geometric reasoning from *osmAG*. The key insight is that encoding wall geometry into signal modeling yields a compact yet generalizable representation: competitive accuracy in data-rich areas and robust extrapolation where purely data-driven methods fail.

Future work will focus on deploying the proposed WiFi+*osmAG* module as a scalable global initializer over large indoor areas, and coupling our system with high-precision LiDAR-based localization [16] to refine the initial global estimate into centimeter-level poses through fine registration. Additionally, we will integrate this WiFi localization module into a complete *osmAG*-based navigation stack [17] [18] to enable end-to-end autonomous operation in large-scale environments.

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